FOURIER TRANSFORM | Quick Reference

Engineering Mathematics · Continuous-Time Angular-Frequency Convention

1. Conceptual Foundation

The **Fourier Transform** represents a (generally aperiodic) time-domain signal $f(t)$ as a continuous spectrum of complex frequency components $F(\omega)$.

Core idea: Any reasonably well-behaved signal can be decomposed into a continuum of complex exponentials $e^{i\omega t}$ (or sines & cosines) of different angular frequencies ω.

"The Fourier Transform represents a signal as a continuous spectrum of frequency components."

Physical meaning: $|F(\omega)|$ = strength at angular frequency ω; $\arg F(\omega)$ = phase.

Relation to Fourier Series: FS applies to *periodic* signals → discrete line spectrum. FT is the continuous analogue for aperiodic/transient signals; series coefficients become a continuous density $F(\omega)$.

Applications: Signal processing & filtering, communications (modulation), vibration analysis, linear PDEs/ODEs, optics, quantum mechanics, medical imaging.

2. Definition & Inverse Transform

Analysis (Forward)

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt$$

Synthesis (Inverse)

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\, e^{i\omega t}\, d\omega$$

- $f(t)$: time-domain signal $F(\omega)$: frequency spectrum
- $\omega = 2\pi f$: angular frequency (rad/s) $i = \sqrt{-1}$

This cheatsheet uses the **angular-frequency, asymmetric** $1/2\pi$ engineering convention. Other normalizations exist; always verify the convention of a given reference.

3. Fundamental Fourier Transform Pairs

Time Domain $f(t)$	$\leftrightarrow$ **Frequency Domain** $F(\omega)$
$\delta(t)$	$\leftrightarrow 1$
1	$\leftrightarrow 2\pi\delta(\omega)$
$e^{-at}u(t)$, $a>0$	$\leftrightarrow \dfrac{1}{a+i\omega}$
$e^{-a\lvert t\rvert}$, $a>0$	$\leftrightarrow \dfrac{2a}{a^2+\omega^2}$
$\cos(\omega_0 t)$	$\leftrightarrow \pi\big[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\big]$
$\sin(\omega_0 t)$	$\leftrightarrow \dfrac{\pi}{i}\big[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\big]$
$\mathrm{rect}(t/T)$	$\leftrightarrow T\ \mathrm{sinc}\big(\frac{\omega T}{2\pi}\big)$

Rect & Sinc convention:

$$\mathrm{rect}(x) = \begin{cases} 1, & |x| \le \frac{1}{2}, \\ 0, & |x| > \frac{1}{2}. \end{cases} \qquad \mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \quad (\mathrm{sinc}(0)=1).$$

Pulse of width T, height $1 \Rightarrow F(\omega) = T\ \mathrm{sinc}(\omega T/2\pi) = 2\sin(\omega T/2)/\omega$.

4. Important Properties

Property	**Time Domain**	**Fourier Transform**
Linearity	$af+bg$	$aF+bG$
Time shift	$f(t-t_0)$	$e^{-i\omega t_0}F(\omega)$
Freq. shift	$e^{i\omega_0 t}f(t)$	$F(\omega-\omega_0)$
Time scale	$f(at)$	$\dfrac{1}{\lvert a\rvert}F(\omega/a)$
Time rev.	$f(-t)$	$F(-\omega)$
Diff. in t	$f'(t)$	$i\omega\, F(\omega)$
Mult. by t	$tf(t)$	$i\dfrac{dF}{d\omega}$
Convolution	$(f*g)(t)$	$F(\omega)G(\omega)$
Product	$f(t)g(t)$	$\dfrac{1}{2\pi}(F*G)(\omega)$

Notes: Scaling uses $|a|$. Differentiation assumes boundary terms vanish. Convolution theorem is the most powerful tool for LTI systems.

5. Conditions of Existence

Sufficient conditions (Dirichlet-type) for the ordinary integral:

- Absolute integrability: $\int_{-\infty}^{\infty} |f(t)|\, dt < \infty$.
- Bounded variation on finite intervals (or piecewise smooth) with finitely many discontinuities.

These are *sufficient*, not necessary. Many practical signals (sinusoids, step, constants) are not absolutely integrable; they possess transforms only as **tempered distributions**. Pairs involving $\delta(\omega)$ are understood distributionally. Engineering practice routinely uses the generalized transforms.

6. Method Selection Guide

Situation	Method	Procedure
Exponential decay	Standard pair	Look up $e^{-at}u(t)$
Time-shifted	Time-shift prop.	Factor $e^{-i\omega t_0}$
Modulated carrier	Freq-shift prop.	Shift spectrum by ω_0
Time-scaled	Scaling prop.	Insert $1/\lvert a\rvert$ & ω/a
Derivative present	Diff. prop.	Multiply by $i\omega$
Convolution of signals	Conv. thm.	Product of spectra
Piecewise definition	Split integral	Integrate on each piece
Periodic signal	Fourier Series	Discrete harmonics only

FT vs. FS: Use continuous FT for aperiodic / transient / finite-energy signals. Use discrete FS for strictly periodic signals (spectrum = impulses at harmonics).

7. Solution Algorithm

1. Identify $f(t)$ and its support (domain).
2. Classify: continuous / piecewise / periodic / exponentially decaying / modulated.
3. Select the matching transform pair or property (Secs. 3–4).
4. If no pair matches, substitute into the defining integral.
5. Evaluate carefully (limits, absolute convergence, distribution sense).
6. Simplify the algebraic expression for $F(\omega)$.
7. Verify: dimensions, even/odd symmetry, known special cases, magnitude decay.

8. Worked Example 1 — Causal Exponential

Problem. Find $\mathcal{F}\{e^{-at}u(t)\}$ for $a > 0$.
Solution.

$$F(\omega) = \int_0^\infty e^{-at}e^{-i\omega t}\,dt = \int_0^\infty e^{-(a+i\omega)t}\,dt$$
$$= \left[-\frac{e^{-(a+i\omega)t}}{a+i\omega}\right]_0^\infty = \frac{1}{a+i\omega}.$$

Result: $F(\omega) = \dfrac{1}{a+i\omega}$.
Magnitude & phase:

$$|F(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}, \qquad \arg F(\omega) = -\tan^{-1}(\omega/a).$$

Interpretation: Low-pass spectrum. Larger a ⇒ shorter time pulse ⇒ broader frequency content (uncertainty principle).

9. Worked Example 2 — Rectangular Pulse

Problem. Let

$$f(t) = \begin{cases} 1, & |t| \le T/2, \\ 0, & |t| > T/2. \end{cases}$$

Derivation.

$$F(\omega) = \int_{-T/2}^{T/2} e^{-i\omega t}\,dt = \left[-\frac{e^{-i\omega t}}{i\omega}\right]_{-T/2}^{T/2}$$
$$= \frac{e^{i\omega T/2} - e^{-i\omega T/2}}{i\omega} = \frac{2\sin(\omega T/2)}{\omega}.$$

At $\omega = 0$ take the limit: $F(0) = T$.
Sinc form: $F(\omega) = T\ \mathrm{sinc}(\omega T/2\pi)$.
Key observations:

- $F(\omega)$ is real and even (f real & even).
- Main-lobe width $\propto 1/T$: shorter pulse ⇒ broader spectrum.
- Side-lobe amplitude decays as $1/|\omega|$.

10. Interpretation & Verification

Spectra: Magnitude $|F(\omega)|$; Phase $\arg F(\omega)$.
Symmetry for real-valued $f(t)$:

$$F(-\omega) = F^*(\omega) \quad \text{(Hermitian)}.$$

⇒ $|F|$ even, $\arg F$ odd.

- Even real f ⇒ F real & even.
- Odd real f ⇒ F purely imaginary & odd.

Parseval / Plancherel (energy):

$$\int_{-\infty}^{\infty} |f(t)|^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega.$$

11. Common Errors — Quick Check

- Confusing ω (rad/s) with ordinary frequency f (Hz). Factor 2π appears/disappears.
- Forgetting the factor $1/(2\pi)$ in the *inverse* transform.
- Sign errors in time-shift $e^{-i\omega t_0}$ and frequency-shift $e^{i\omega_0 t}$.
- Omitting the absolute value $|a|$ in the time-scaling property.
- Incorrect limits for piecewise signals; always split the integral.
- Applying continuous FT to a strictly periodic signal (use FS instead).
- Treating δ-function pairs as ordinary functions.
- Inconsistent sinc normalization (normalized vs. unnormalized).
- Evaluating $2\sin(\omega T/2)/\omega$ by direct substitution at $\omega = 0$ instead of the limit.